

Class 5: Data Viz with ggplot2

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Background

There are many graphics systems in R for making plots and figures. These include so-called *base R graphics* like the `plot()` function and add on packages like **ggplot2**.

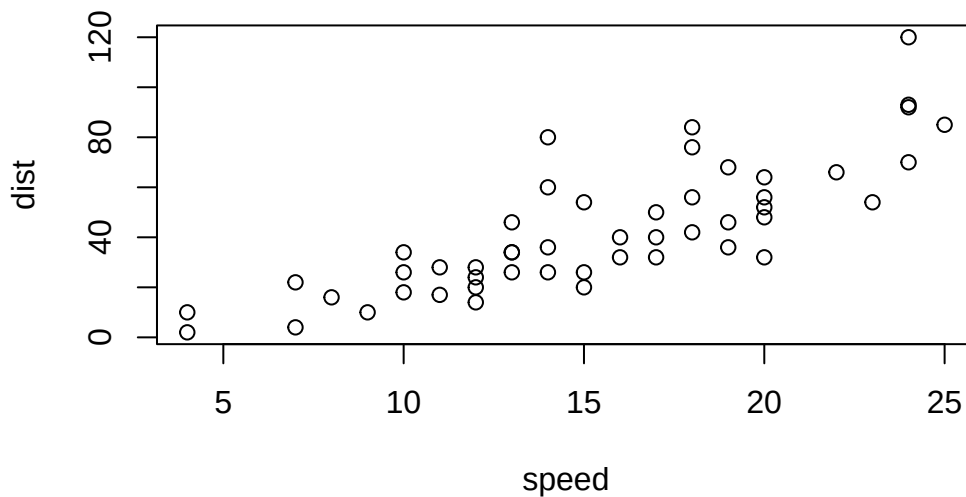
Let's compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use `ggplot2`, I need to install it on my computer. To do this, we can use the function `install.packages("ggplot2")`

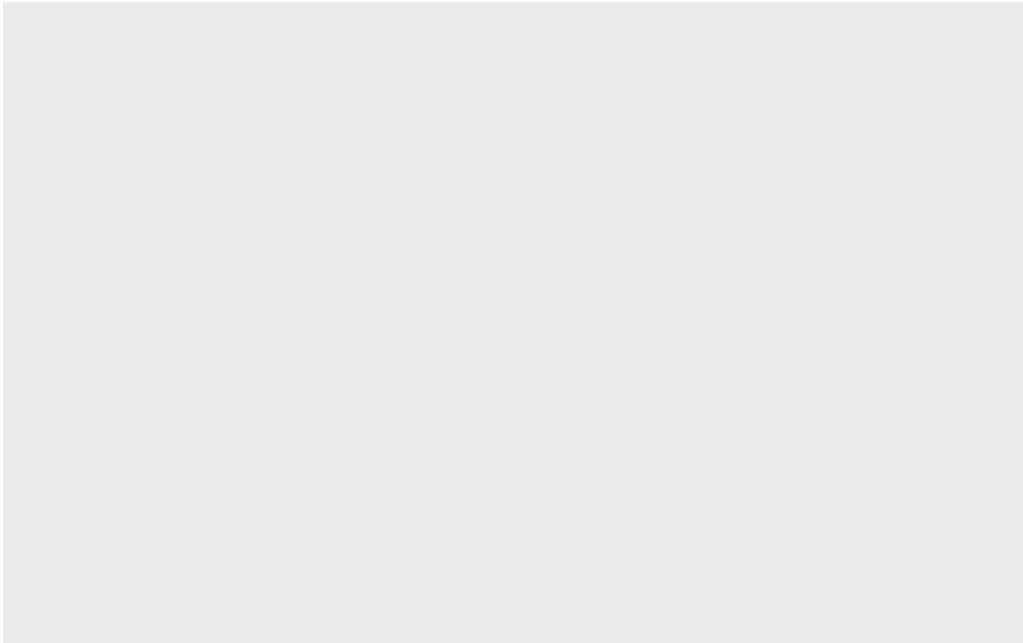
N.B. We never run `install.packages()` in our quarto doc (we only run it once in our R console) as it would re-install the package every time we render our quarto report.

Once installed, we need to load up the package into our R brain:

```
library(ggplot2)
```

The main function in the **ggplot2** package is called `ggplot()`

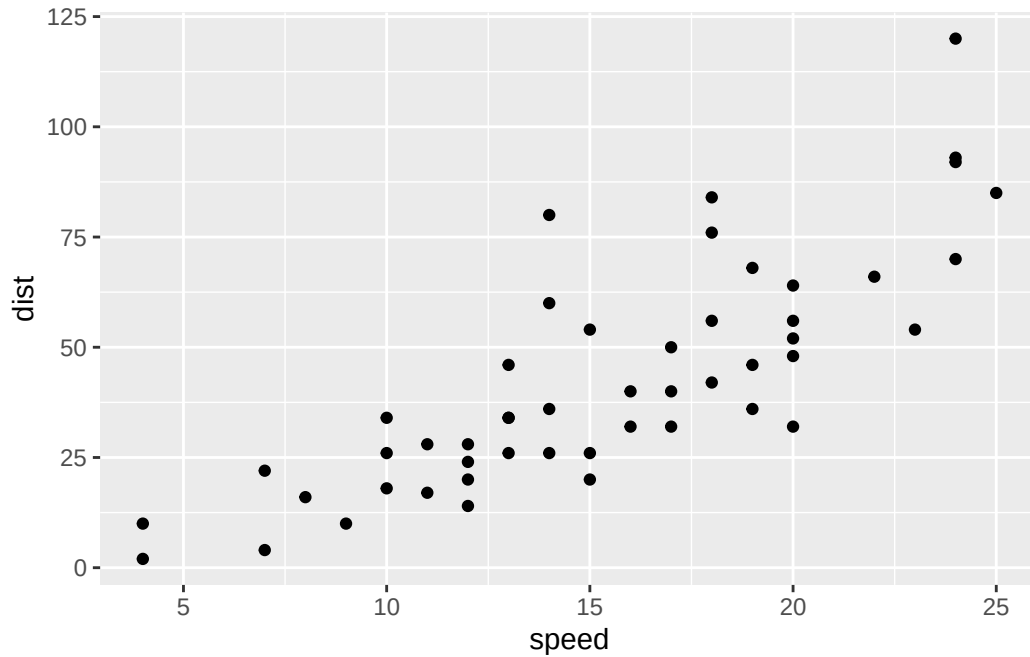
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- First Layer: the **data** (a data.frame of the stuff we want to plot)
- Second Layer: the **aesthetics** (how the data maps to the plot)
- Third Layer: the **geom** layer (how you want the plot drawn, e.g. points, lines, bars, etc.)

```
ggplot(cars) +  
  aes(x = speed, y = dist) +  
  geom_point()
```

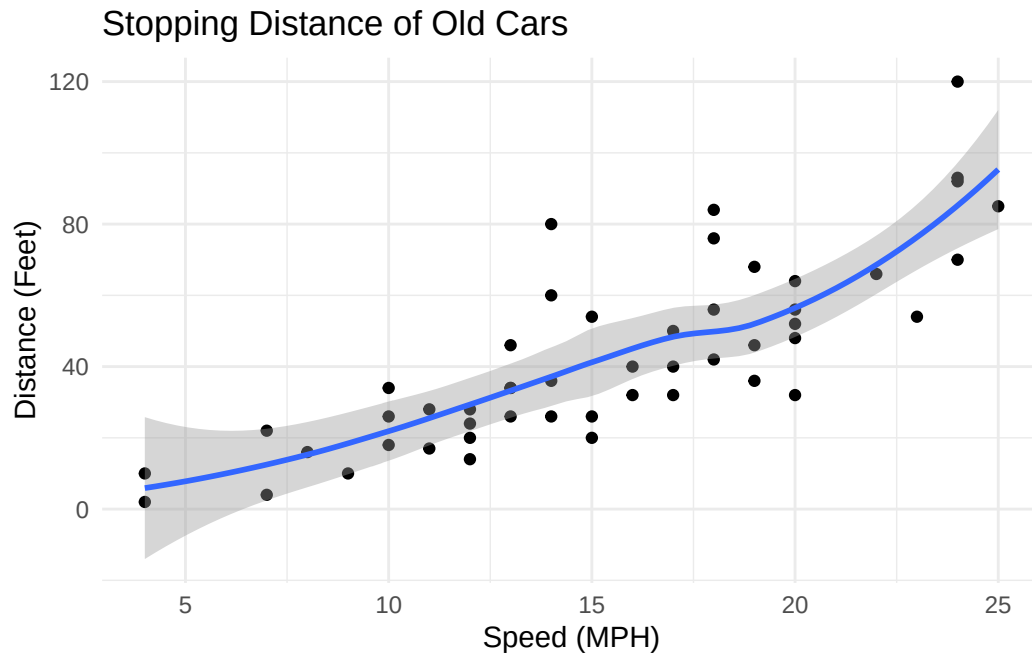


Add some custom features to the plot

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x = speed, y = dist) +  
  geom_point() +  
  geom_smooth() +  
  theme_minimal() +  
  labs(title = "Stopping Distance of Old Cars", x = "Speed (MPH)", y = "Distance (Feet)")
```

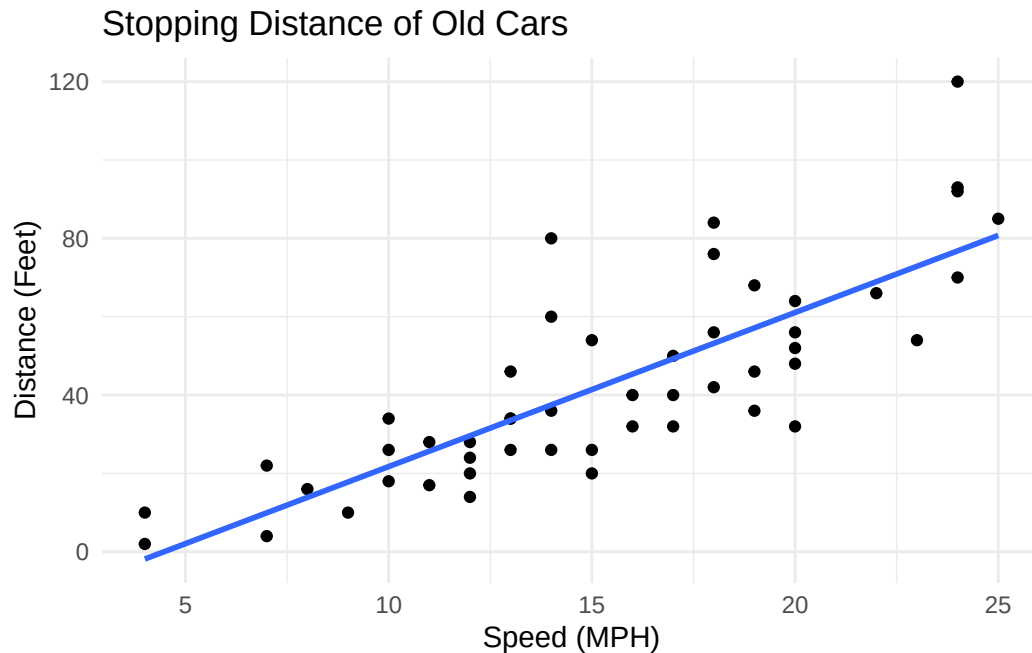
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



Q. Can you make the `geom_smooth()` function produce a linear line of best fit for the data and turn off the “gray” error region?

```
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal() +
  labs(title = "Stopping Distance of Old Cars", x = "Speed (MPH)", y = "Distance (Feet)")
```

`geom_smooth()` using `formula = 'y ~ x'`



Gene Expression Figure

Import the data to plot.

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
sum(genes$State == "up")
```

```
[1] 127
```

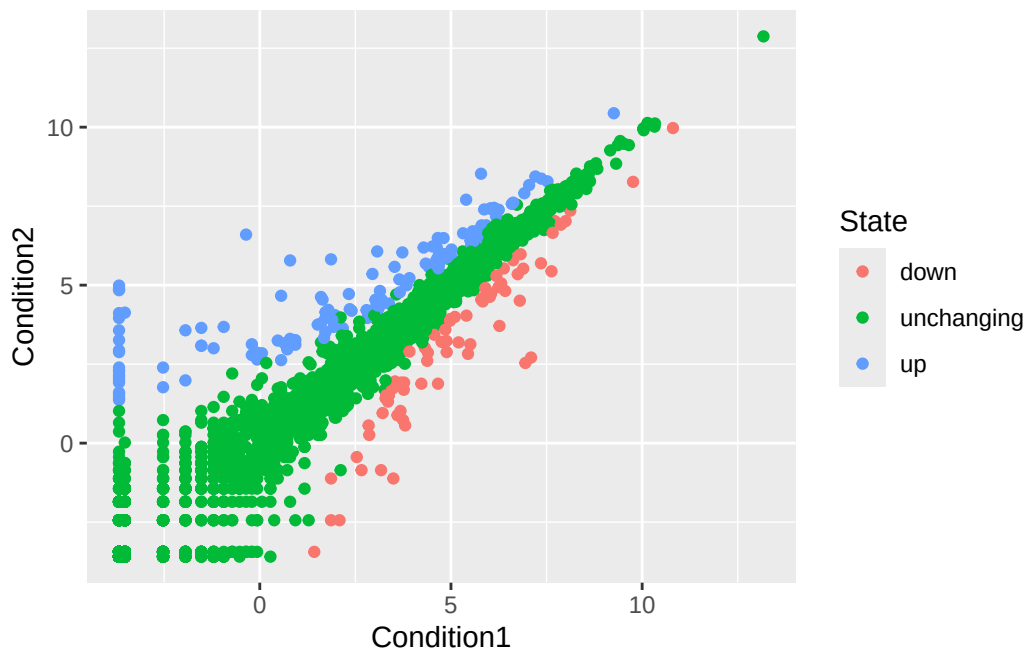
A useful new function in this context is the `table()` function:

```
table(genes$State)
```

down	unchanging	up
72	4997	127

Initial Plot

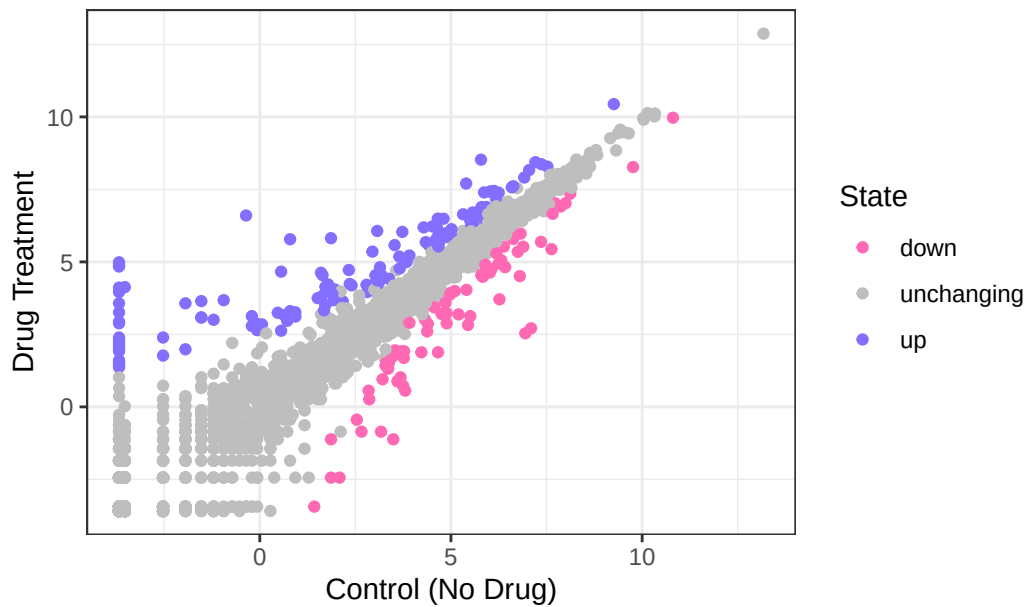
```
ggplot(genes) +  
  aes(Condition1, Condition2, col = State) +  
  geom_point()
```



Modified Plot

```
ggplot(genes) +  
  aes(Condition1, Condition2, col = State) +  
  geom_point() +  
  theme_bw() +  
  labs(title = "Gene Expression Changes Upon GLP-1 Inhibitor Treatment", x = "Control (No Drug)", y = "GLP-1 Inhibitor") +  
  scale_color_manual(values = c("hotpink", "gray", "slateblue1"))
```

Gene Expression Changes Upon GLP-1 Inhibitor Treatment



Going Further

Here we read the famous gapminder dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different “country” entries are in this dataset?

```
length(table(gapminder$country))
```

```
[1] 142
```

```
length(unique(gapminder$country))
```

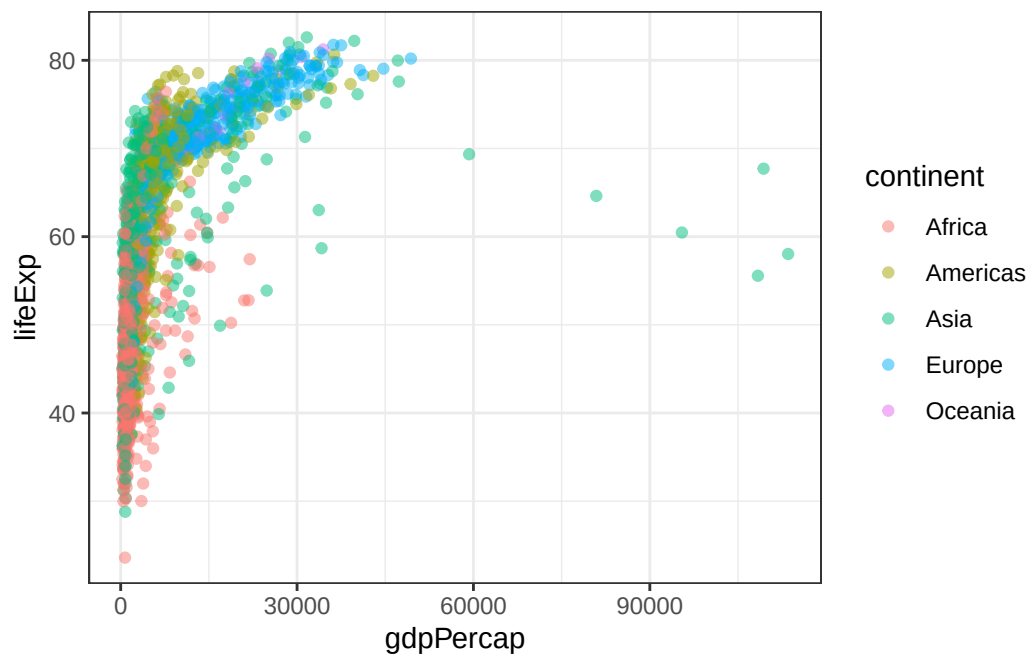
```
[1] 142
```

Let’s make our first plot of the entire dataset:

Plot of “gdpPercap” vs “lifeExp” colored by “continent”

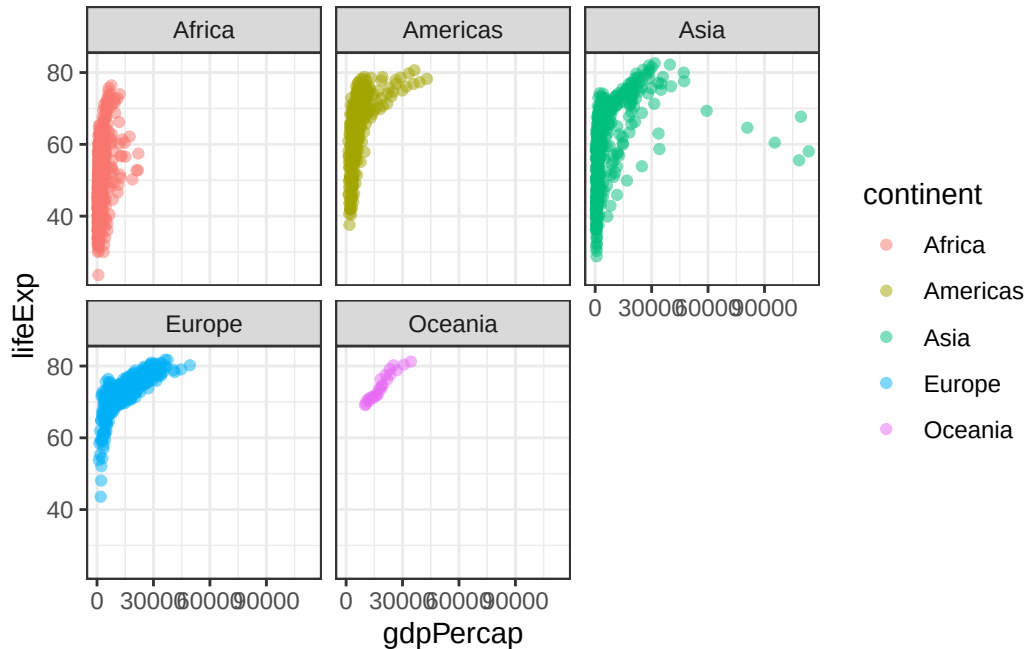
```
p <- ggplot(gapminder) +  
  aes(x = gdpPercap, y = lifeExp, col = continent) +  
  geom_point(alpha=0.5) +  
  theme_bw()
```

p



I can add more layers to `p`, which is my base plot.

```
p +  
  facet_wrap(~gapminder$continent)
```



Make a plot for 1977 and 2007 only (not all the years in the dataset).

First use the **dplyr** package and the `filter()` function from that package to extract the year 2007.

```
library(dplyr)
```

Attaching package: 'dplyr'

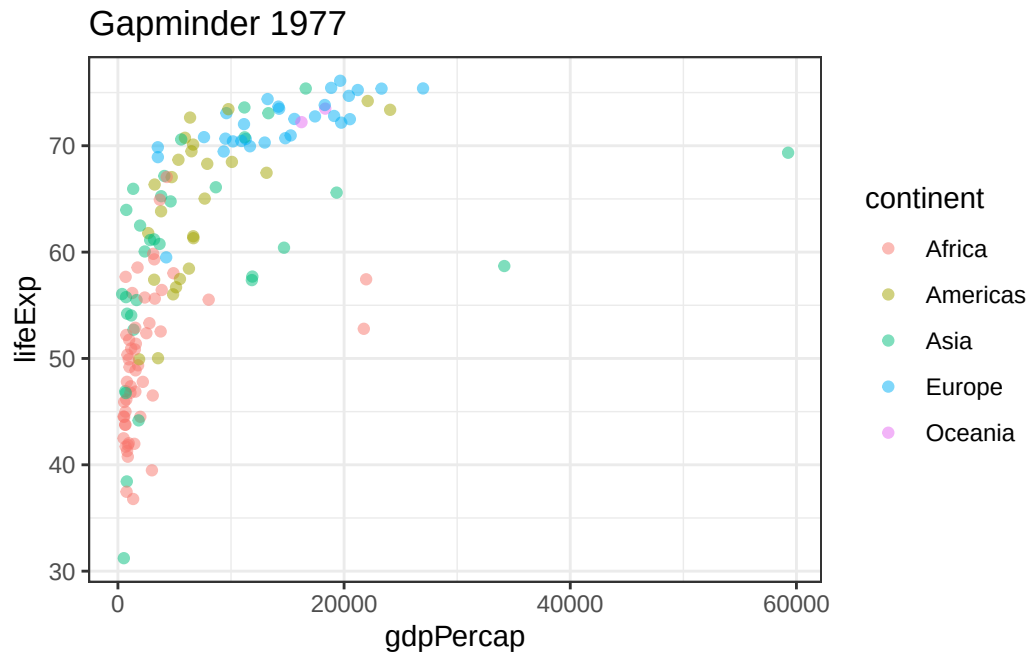
The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

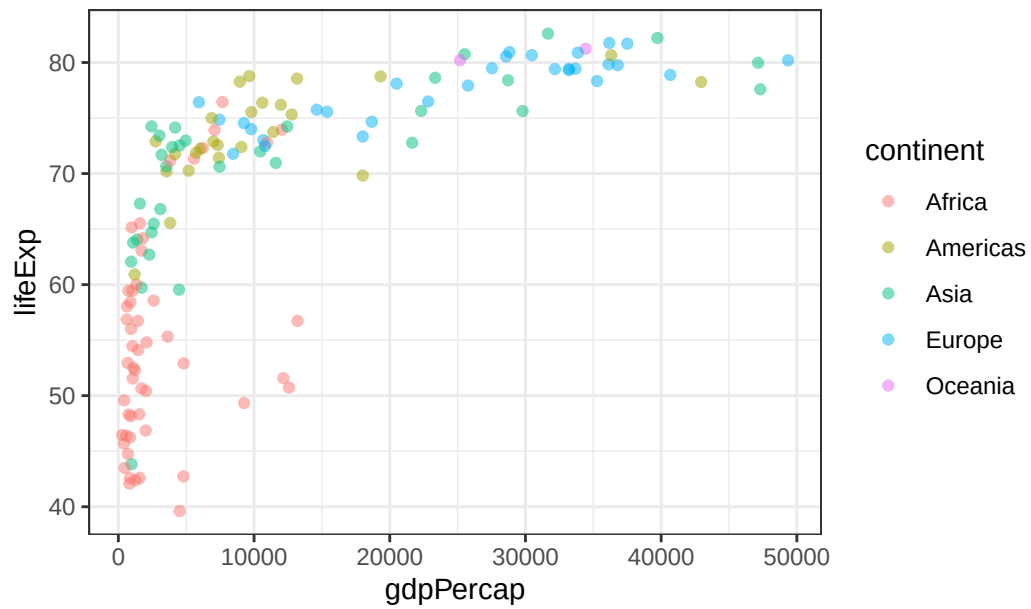
`intersect`, `setdiff`, `setequal`, `union`

```
gapminder_1977 <- filter(gapminder, year==1977)
ggplot(gapminder_1977) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point(alpha=0.5) +
  theme_bw() +
  labs(title = "Gapminder 1977")
```



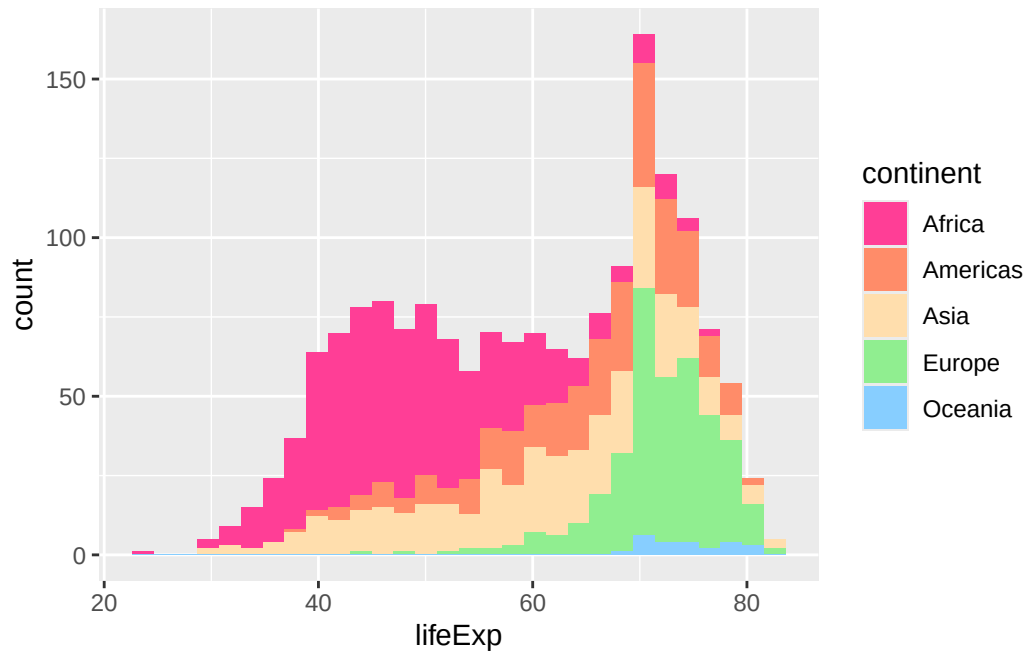
```
gapminder_2007 <- filter(gapminder, year==2007)
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point(alpha=0.5) +
  theme_bw() +
  labs(title = "Gapminder 2007")
```

Gapminder 2007



Q. Make a histogram of lifeExp colored by continent

```
ggplot(gapminder, aes(x = lifeExp, fill = continent)) +  
  geom_histogram(bins = 30) +  
  scale_fill_manual(values = c("violetred1", "salmon1", "navajowhite1", "lightgreen", "skyblu
```



Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder, aes(x = lifeExp)) +  
  geom_histogram(bins = 30) +  
  facet_wrap(~ continent)
```

